

The 168 Count for Octonion Basis Associators and a Computational Cayley–Dickson Tower Extension

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Abstract. We prove that the number of ordered triples of imaginary octonion basis elements with nonzero associator is exactly $168 = |\mathrm{PSL}(2,7)|$. The result follows from the regular action of $\mathrm{Aut}(\mathrm{Fano}) \cong \mathrm{PGL}(3, \mathbb{F}_2) \cong \mathrm{PSL}(2, 7)$ on ordered non-collinear triples. We observe that the associator norm on basis elements takes exactly the values $\{0, 2\}$ — a $\mathbb{Z}_2$ dichotomy with no intermediate magnitudes — and prove this from the Cayley–Dickson $\mathbb{Z}_2^k$ grading without case analysis. Exhaustive computation over the sedenion ($\dim = 16$) and trigintaduonion ($\dim = 32$) algebras reveals that all nonzero basis associator counts are multiples of 168: $1848 = 11 \times 168$ for sedenions and $15\,960 = 95 \times 168$ for trigintaduonions. The norm dichotomy extends to both higher dimensions. We conjecture a closed formula $T_k = 168(P_k - 4P_{k-1})$, where P_k counts Fano subplanes in $\mathrm{PG}(k-1, 2)$, verified at $k = 3, 4, 5$. The primitive zero-divisor pair count $336 = 2 \times 168$ in sedenions is shown to be a combinatorial artifact of the $\mathbb{Z}_2^4$ upper-half structure, not a structural bijection. All enumerations are verified exhaustively in the Sounio programming language.

Keywords: octonions, sedenions, trigintaduonions, associator, Fano plane, $\mathrm{PSL}(2,7)$, Cayley–Dickson, zero divisors, projective geometry

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1. Introduction

The Cayley–Dickson construction produces a tower of algebras by iterated doubling:

$$\mathbb{R} \xrightarrow{\times 2} \mathbb{C} \xrightarrow{\times 2} \mathbb{H} \xrightarrow{\times 2} \mathbb{O} \xrightarrow{\times 2} \mathbb{S} \xrightarrow{\times 2} \mathbb{T} \xrightarrow{\times 2} \dots \quad (1)$$

At each step, a structural property is lost: commutativity at the quaternions $\mathbb{H}$, associativity at the octonions $\mathbb{O}$, and the division property at the sedenions $\mathbb{S}$ (which acquire zero divisors) [1, 2]. The octonions are the last normed division algebra (Hurwitz, 1898) and have found applications in string theory [3], exceptional Lie groups [4], and quantum information [5].

The multiplication table of the octonions is encoded by the Fano plane — the unique projective plane of order 2, with 7 points and 7 lines. The automorphism group of the Fano plane is $\mathrm{Aut}(\mathrm{Fano}) \cong \mathrm{PGL}(3, \mathbb{F}_2) = \mathrm{GL}(3, \mathbb{F}_2) \cong \mathrm{PSL}(2, 7)$, a simple group of order 168 [6]. This group acts on the octonion basis elements by permuting them while preserving the multiplication structure.

While the role of $\mathrm{PSL}(2, 7)$ in octonion multiplication has been extensively studied [1, 4, 7], we provide an explicit combinatorial treatment of the basis associator count and extend it computationally up the Cayley–Dickson tower:

The number of ordered triples $(i, j, k) \in \{1, \dots, 7\}^3$ for which the octonion basis associator $[e_i, e_j, e_k] = (e_i e_j) e_k - e_{i(e_j e_k)}$ is nonzero is exactly 168. This count is a multiple of 168 in every higher Cayley–Dickson algebra tested.

We prove the octonionic result, establish a norm dichotomy for basis associators (with a proof avoiding case analysis), and report exhaustive computations extending the 168-divisibility pattern to sedenions ($\dim = 16$) and trigintaduonions ($\dim = 32$).

Contributions. (1) *Theorem 1:* the nonzero octonion basis associator count equals $|\mathrm{PSL}(2, 7)|$, with decomposition $343 = 133 + 42 + 168$. (2) *Lemma 2:* $\|[e_i, e_j, e_k]\| \in \{0, 2\}$ for all basis elements, with a proof via the $\mathbb{Z}_2^k$ grading of the Cayley–Dickson construction. (3) *Computational observations* (Sections 4–5): all nonzero associator counts in sedenions and trigintaduonions are multiples of 168; the norm dichotomy extends to $\dim = 32$. (4) *Conjecture 5:* a closed formula $T_k = 168(P_k - 4P_{k-1})$ for the nonzero associator count in terms of Fano subplane counts.

2. The 168 Theorem

2.1. Setup

Let $\{e_0, e_1, \dots, e_7\}$ denote the standard octonion basis, where $e_0 = 1$ is the identity and $e_1, \dots, e_7$ are imaginary units. The multiplication is determined by the Fano plane: for each oriented line (i, j, k) , we have $e_i e_j = e_k$ and $e_j e_i = -e_k$.

The **associator** of three octonions is:

$$[a, b, c] = (ab)c - a(bc) \quad (2)$$

The octonions are *alternative*: $[a, a, b] = [a, b, b] = 0$ for all a, b . By the Artin theorem, any subalgebra generated by two elements is associative [1].

2.2. Counting nonzero basis associators

We restrict to imaginary basis elements $e_1, \dots, e_7$, excluding the real unit e_0 (for which all associators vanish trivially). Consider all $7^3 = 343$ ordered triples $(i, j, k) \in \{1, \dots, 7\}^3$. We use ordered triples because the associator is alternating: $[a, b, c] = -[b, a, c]$; the corresponding count of *unordered* non-collinear triples is $168/6 = 28$.

Layer 1: Repeated indices (133 zeros). If any two of $\{i, j, k\}$ coincide, the associator vanishes by alternativity ($[a, a, b] = 0$) or flexibility ($[a, b, a] = 0$, which holds in any alternative algebra [1]). By inclusion-exclusion over the conditions $i = j$, $j = k$, $i = k$:

$$3(49) - 3(7) + 7 = 133 \quad (3)$$

Layer 2: Fano-line triples (42 zeros). Among the $7 \times 6 \times 5 = 210$ triples with all-distinct indices, those where $\{i, j, k\}$ forms a line of the Fano plane generate a quaternion subalgebra, which is associative. There are 7 lines, each contributing $3! = 6$ ordered triples: $7 \times 6 = 42$.

Layer 3: Non-collinear triples (168 nonzero). The remaining: $343 - 133 - 42 = 168$.

2.3. The group-theoretic explanation

Theorem 1. *The number of nonzero basis associator triples equals $|\mathrm{PSL}(2, 7)| = 168$.*

Proof. The Fano plane $\mathbb{F}$ has 7 points and 7 lines. An ordered non-collinear triple is an ordered triple (p, q, r) of distinct points not all on a line. The count is $7 \times 6 \times 5 - 7 \times 6 = 168$.

$\mathrm{Aut}(\mathbb{F}) \cong \mathrm{PGL}(3, \mathbb{F}_2) = \mathrm{GL}(3, \mathbb{F}_2) \cong \mathrm{PSL}(2, 7)$ acts on ordered non-collinear triples. This action is **regular** (free and transitive):

- *Transitive:* $\mathrm{GL}(3, \mathbb{F}_2)$ acts transitively on ordered bases of $\mathbb{F}_2^3$. Non-collinear triples in $\mathbb{F}$ correspond to ordered bases of $\mathbb{F}_2^3$ under the identification $\mathbb{F} \cong \mathbb{P}(\mathbb{F}_2^3)$, since three projective points are non-collinear if and only if the corresponding vectors are linearly independent.

- *Free*: The stabilizer of an ordered basis of $\mathbb{F}_2^3$ in $\text{GL}(3, \mathbb{F}_2)$ is trivial, since an invertible linear map is uniquely determined by its action on a basis.

Therefore $|\{\text{nonzero associator triples}\}| = |\text{PSL}(2, 7)| = 168$. $\square$

3. Basis Associator Norm Dichotomy

Lemma 2. For all $i, j, k \in \{1, \dots, 7\}$: $\| [e_i, e_j, e_k] \| \in \{0, 2\}$.

Proof. For distinct imaginary basis elements e_i, e_j ($i \neq j$, both ≥ 1), the Fano plane multiplication gives $e_i e_j = \epsilon_{ij} e_{f(i,j)}$ where $\epsilon_{ij} \in \{+1, -1\}$ and $f(i, j) \in \{1, \dots, 7\}$. Therefore:

$$(e_i e_j) e_k = \epsilon_{ij} \cdot \epsilon_{f(i,j), k} \cdot e_{f(f(i,j), k)} = \pm e_p \quad (4)$$

$$e_i (e_j e_k) = \epsilon_{jk} \cdot \epsilon_{i, f(j,k)} \cdot e_{f(i, f(j,k))} = \pm e_q \quad (5)$$

for some $p, q \in \{0, 1, \dots, 7\}$ (where index 0 arises when both factors are equal, giving $e_m^2 = -e_0$, a real scalar). Each parenthesization yields $\pm$ a single basis element.

We claim $p = q$, i.e., both parenthesizations land on the same basis element (possibly with different signs). This follows from the $\mathbb{Z}_2^k$ grading of the Cayley–Dickson construction: there exists a labeling of the basis in which the product $e_i e_j$ has index $i \oplus j$ (bitwise exclusive-or over $\mathbb{Z}_2^3$), up to sign. Since $\oplus$ is associative on $\mathbb{Z}_2^3$:

$$(i \oplus j) \oplus k = i \oplus (j \oplus k) \quad (6)$$

so both parenthesizations yield the same index. The assertion $p = q$ is a property of the abstract algebra $\mathbb{O}$, independent of the choice of labeling: all valid octonion multiplication tables define isomorphic algebras, so a labeling-independent property that holds in one labeling holds in all.

Since $p = q$, the associator $[e_i, e_j, e_k] = (\pm 1 \mp 1) e_p$, which is either 0 (signs match) or $\pm 2e_p$ (signs oppose, giving $\| \pm 2e_p \| = 2$). $\square$

Remark. Non-associativity for basis elements is a $\mathbb{Z}_2$ parity phenomenon — a sign flip within a single basis direction, not a rotation to a different direction. The $\mathbb{Z}_2^k$ grading argument for $p = q$ extends to all Cayley–Dickson algebras, since the index relation $f(i, j) = i \oplus j$ holds universally in the standard Cayley–Dickson construction.

4. Computational Extension to Higher Dimensions

We extend the analysis computationally to the sedenions $\mathbb{S}$ (dim = 16) and tringtaduonions $\mathbb{T}$ (dim = 32), constructed by iterated Cayley–Dickson doubling.

Algebra	Dim	Total triples	Nonzero	Factor
Octonion $\mathbb{O}$	8	$7^3 = 343$	168	1×168
Sedenion $\mathbb{S}$	16	$15^3 = 3375$	1848	11×168
Tringtaduonion $\mathbb{T}$	32	$31^3 = 29\,791$	15\,960	95×168

Table 1: Nonzero basis associator counts across the Cayley–Dickson tower (exhaustive computation). Every entry is a multiple of 168.

Observation 3. Every nonzero associator count in Table 1 is a multiple of 168. The norm dichotomy ($\| [e_i, e_j, e_k] \| \in \{0, 2\}$) extends to all sedenion and tringtaduonion basis elements.

These are computational observations verified by exhaustive enumeration over all $7^3 = 343$, $15^3 = 3375$, and $31^3 = 29\,791$ ordered basis triples in dimensions 8, 16, and 32 respectively.

4.1. Sedenion sub-class decomposition

Sub-class	Range	Total	Nonzero	Factor
Oct-oct-oct	$i, j, k \in \{1, \dots, 7\}$	343	168	1×168
Sed-sed-sed	$i, j, k \in \{8, \dots, 15\}$	512	336	2×168
Cross (mixed)	at least one from each half	2520	1344	8×168
Total	$i, j, k \in \{1, \dots, 15\}$	3375	1848	11×168

Table 2: Sedenion sub-class decomposition. Every entry is a multiple of 168.

The sed-sed count $336 = 2 \times 168$ has a simple explanation: in the $\mathbb{Z}_2^4$ representation, the upper-half indices $\{8, \dots, 15\}$ all share bit 3. For any two upper-half indices a, b : bit 3 of $a \oplus b$ is $1 \oplus 1 = 0$, so $a \oplus b$ falls in the lower half. Therefore no line of $\text{PG}(3, 2)$ (consisting of $\{a, b, a \oplus b\}$) lies entirely in the upper half, and all $8 \times 7 \times 6 = 336$ ordered triples of distinct upper-half elements are non-collinear — hence non-associating. This argument does not extend to zero-divisor structure; at $\dim = 32$ the analogous upper-half count is $16 \times 15 \times 14 = 3360$, breaking the numerical coincidence (see §5.2).

5. Primitive Zero-Divisor Enumeration

The sedenions are the first Cayley–Dickson algebra with zero divisors [2, 8].

5.1. Restricted enumeration

We enumerate zero-divisor pairs of the restricted form $z = e_i + s_1 e_j$, $w = e_k + s_2 e_l$ (two-term basis sums with $\pm$ signs) — the “primitive unit zero-divisors” in the terminology of de Marrais [9]. The full set of sedenion zero-divisors is larger (including arbitrary linear combinations); our enumeration covers only this algebraically fundamental subclass.

Exhaustive search over all such pairs with $i < j$, $k < l$, $s_1, s_2 \in \{+1, -1\}$, and $i, j, k, l \in \{0, \dots, 15\}$ yields exactly **336** ordered pairs (84 unordered pairs $\times$ 4 sign choices). The first pair found is $(e_1 + e_{10})(e_3 + e_{14}) = 0$.

$$336 = 2 \times 168 = 2 \times |\text{PSL}(2, 7)| \quad (7)$$

5.2. Resolution of the $336 = 336$ coincidence

Observation 4. *The primitive zero-divisor pair count (336) equals the sed-sed nonzero associator count (336, Table 2 row 2). Both are $2 \times |\text{PSL}(2, 7)|$.*

This numerical equality is a combinatorial artifact, not a structural bijection. The sed-sed count $336 = 8 \times 7 \times 6$ follows from the $\mathbb{Z}_2^4$ upper-half structure (§4.2): every ordered triple of distinct upper-half elements is non-collinear. The zero-divisor count 336 has an independent origin in the structure constants of the Cayley–Dickson product. At dimension 32, the upper-half count becomes $16 \times 15 \times 14 = 3360$, while the zero-divisor structure scales differently — confirming that the $336 = 336$ equality is particular to the sedenions.

de Marrais [9] independently counted 168 “primitive unit zero-divisors” in sedenions, arranged in 42 “Assessors”; our 336 counts *ordered* pairs ($336 = 2 \times 168$), consistent with his enumeration.

6. Computational Verification

All enumerations are verified by exhaustive computation in the Sounio programming language [10], with native Cayley–Dickson algebra support:

- Octonion multiplication via Fano plane triples $(1,2,4)(2,3,5)(3,4,6)(4,5,7)(5,6,1)(6,7,2)(7,1,3)$.
- Sedenion and trigtaduonion multiplication via iterated Cayley–Dickson: $(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c})$.
- All $7^3 = 343$ octonion, $15^3 = 3375$ sedenion, and $31^3 = 29\,791$ trigtaduonion associator norms computed and classified.
- All 57,120 candidate zero-divisor pairs tested.

- A related result — path product norm invariance for octonion-labeled graphs (a consequence of norm multiplicativity applied to binary tree evaluations) — is machine-checked in Lean 4 with 0 sorry statements.

Source code: github.com/agourakis82/sounio.

7. Discussion

7.1. The orbit size 168 in the Cayley–Dickson tower

The number 168 appears at multiple points in the Cayley–Dickson tower: (1) as the count of nonzero octonion basis associators (Theorem 1); (2) as the denominator in Wilmot’s formula $T_n = (2^n - 1)(2^n - 2)(2^n - 4)/168$ for counting Cayley–Dickson automorphisms [11] (3) as one-half the primitive sedenion zero-divisor pair count; (4) as the universal divisor of nonzero associator counts across the tower (Table 1). These appearances confirm that $|\mathrm{PSL}(2, 7)| = 168$ serves as a basic orbit size in the combinatorics of the tower.

7.2. A conjectured tower formula

Let P_k denote the number of Fano subplanes in the projective geometry $\mathrm{PG}(k - 1, 2)$:

$$P_k = \frac{(2^k - 1)(2^{k-1} - 1)(2^{k-2} - 1)}{21} \quad (8)$$

with $P_2 = 0$, $P_3 = 1$, $P_4 = 15$, $P_5 = 155$.

Conjecture 5. *For all $k \geq 3$, the number of ordered basis triples with nonzero associator in the 2^k -dimensional Cayley–Dickson algebra is:*

$$T_k = 168(P_k - 4P_{k-1}) \quad (9)$$

Verification. The conjecture is confirmed computationally at three levels:

- $k = 3$ (octonions): $T_3 = 168(1 - 0) = 168$ ✓ (Theorem 1)
- $k = 4$ (sedenions): $T_4 = 168(15 - 4) = 1848$ ✓ (Observation 3)
- $k = 5$ (trigintaduonions): $T_5 = 168(155 - 60) = 15\,960$ ✓ (new computation)

The formula predicts $T_6 = 168(1395 - 620) = 130\,200$ for the 64-dimensional Cayley–Dickson algebra.

Remark. An analytical proof may follow from the decomposition of Fano subplanes in $\mathrm{PG}(k - 1, 2)$ into “octonionic” subplanes (preserving alternativity, contributing 168 nonzero triples each) and “quasi-octonionic” subplanes (where alternativity fails, contributing fewer), as studied by Cawagas [8] for the sedenion case. We leave the general proof as an open problem.

Remark. The factors 1, 11, 95 are given by $P_k - 4P_{k-1}$. In particular, the factor $11 = P_4 - 4P_3 = 15 - 4$ in $1848 = 11 \times 168$ has combinatorial rather than group-theoretic significance (note $11 \nmid 168 = 2^3 \cdot 3 \cdot 7$).

7.3. Norm dichotomy

The $\mathbb{Z}_2$ character of basis non-associativity (Lemma 2) — a sign flip, not a continuous rotation — implies that for basis elements, non-associativity is a discrete phenomenon. This extends computationally to sedenions and trigintaduonions (Observation 3), suggesting it is a general property of the Cayley–Dickson construction, derivable from the bilinearity of the structure constants, the ± 1 character of the signs, and the $\mathbb{Z}_2^k$ index grading.

7.4. Open questions

The original open questions from the initial submission are now resolved (OQ1: $168 \times$ divisibility confirmed at $\dim = 32$; OQ2: $336 = 336$ explained as combinatorial artifact; OQ3: factor 11 identified via P_k formula; OQ4: $p = q$ proved via $\mathbb{Z}_2^k$ grading). The following questions remain:

1. Prove Conjecture 5 analytically for all $k \geq 3$. The key step would be establishing the octonionic/quasi-octonionic subplane decomposition and showing that quasi-octonionic planes contribute exactly 72 nonzero associator triples.

2. Does the norm dichotomy $\|[e_i, e_j, e_k]\| \in \{0, 2\}$ hold for *all* Cayley–Dickson algebras ($k \geq 6$)? It is verified at $k = 3, 4, 5$.
3. Characterize the sub-class distribution (analogous to Table 2) for trigintaduonions and higher, and determine whether the sub-class counts are individually multiples of 168.

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